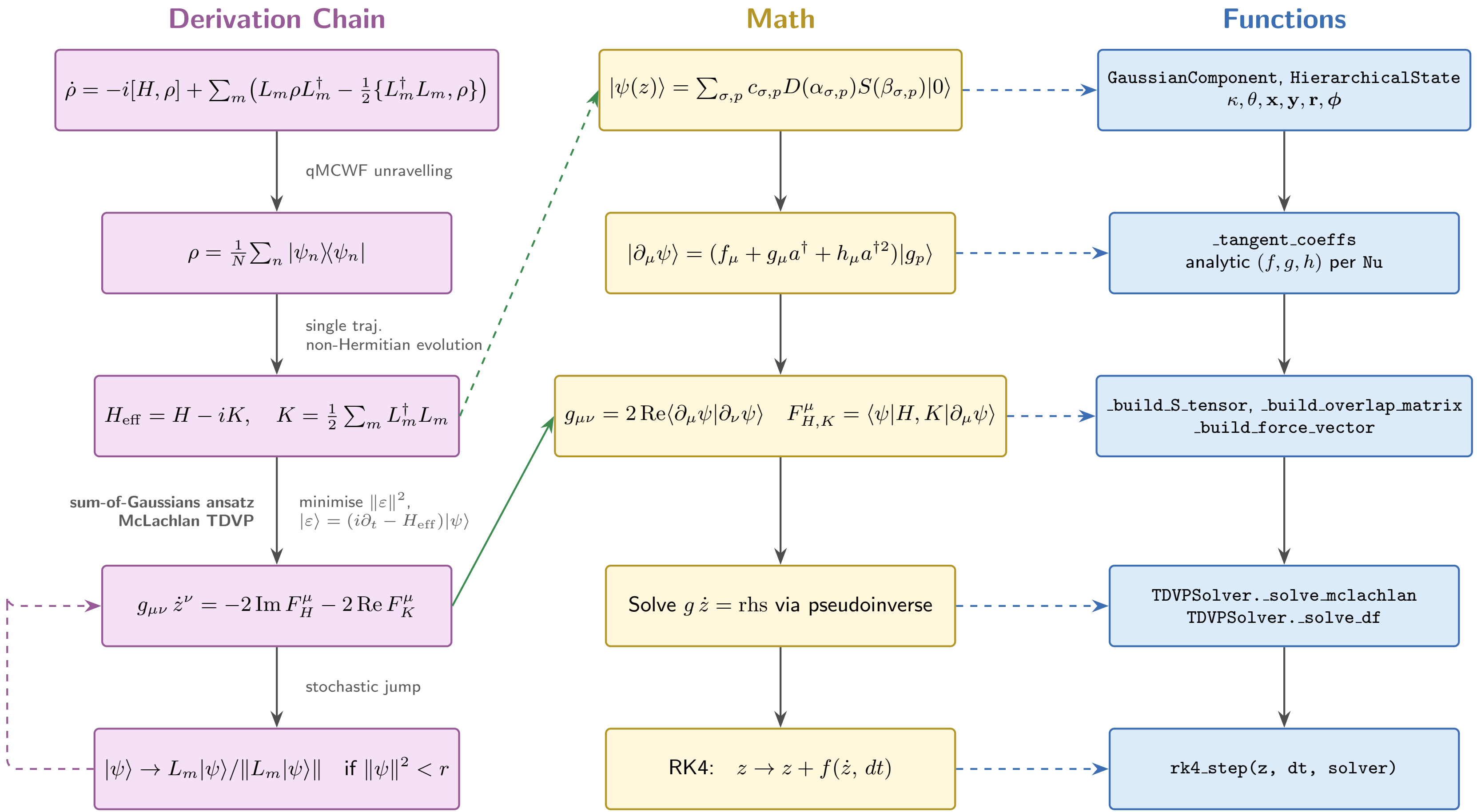
Derivation Chain
Math
Functions
$\dot\rho = -i[H,\rho] + \sum_m \left(L_m \rho L_m^\dagger - \frac{1}{2}\{L_m^\dagger L_m, \rho\}\right)$
qMCWF unravelling
$\rho = \frac{1}{N}\sum_n |\psi_n\rangle\langle\psi_n|$
single traj.
non-Hermitian evolution
$H_{\text{eff}} = H - iK, \quad K = \frac{1}{2}\sum_m L_m^\dagger L_m$
sum-of-Gaussians ansatz
McLachlan TDVP
minimise $\|\varepsilon\|^2$,
$|\varepsilon\rangle = (i\partial_t - H_{\text{eff}})|\psi\rangle$
$g_{\mu\nu}\,\dot z^\nu = -2\,\mathrm{Im}\,F_H^\mu - 2\,\mathrm{Re}\,F_K^\mu$
stochastic jump
$|\psi\rangle \to L_m|\psi\rangle/\|L_m|\psi\rangle\|$ if $\|\psi\|^2 < r$
$|\psi(z)\rangle = \sum_{\sigma,p} c_{\sigma,p} D(\alpha_{\sigma,p}) S(\beta_{\sigma,p})|0\rangle$
$|\partial_\mu\psi\rangle = (f_\mu + g_\mu a^\dagger + h_\mu a^{\dagger 2})|g_p\rangle$
$g_{\mu\nu} = 2\,\mathrm{Re}\langle\partial_\mu\psi|\partial_\nu\psi\rangle \quad F^\mu_{H,K} = \langle\psi|H,K|\partial_\mu\psi\rangle$
Solve $g\,\dot z = \mathrm{rhs}$ via pseudoinverse
RK4: $z \to z + f(\dot z,\, dt)$
GaussianComponent, HierarchicalState
$\kappa, \theta, \mathbf{x}, \mathbf{y}, \mathbf{r}, \boldsymbol{\phi}$
_tangent_coeffs
analytic (f, g, h) per Nu
_build_S_tensor, _build_overlap_matrix
_build_force_vector
TDVPSolver._solve_mclachlan
TDVPSolver._solve_df
rk4_step(z, dt, solver)